Drone Technology and its Transformative Applications

TOPIC-Urban Change Detection- Encroachment Monitoring

Project report submitted in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology
in
Robotics and Artificial Intelligence (2nd Year)

BY

RUDRA SINGH	**(2410090009)** Text
SAMDISH VASHISTHA	**(2410090010)**
SARVAGYA SHUKLA	**(2410090011)**
SHISHIR GUPTA	**(2410090012)**

IILM UNIVERSITY

Github link-https://github.com/sarvagyashuklara28-netizen/Urban-Change-Dectection

Department of Electrical and Electronics Engineering
School of Engineering, IILM University, Greater Noida
Gautam Buddha Nagar, Uttar Pradesh

APRIL 2026 (2025-26)

Urban Change Detection – Encroachment Monitoring

Aim

The aim of this project is to detect urban land-cover changes and monitor encroachment using aerial or satellite images.

Objectives

- Segment aerial images into different land-cover categories using OpenCV.
- Detect urban changes and encroachment.
- Calculate area percentages.
- Generate segmented outputs.

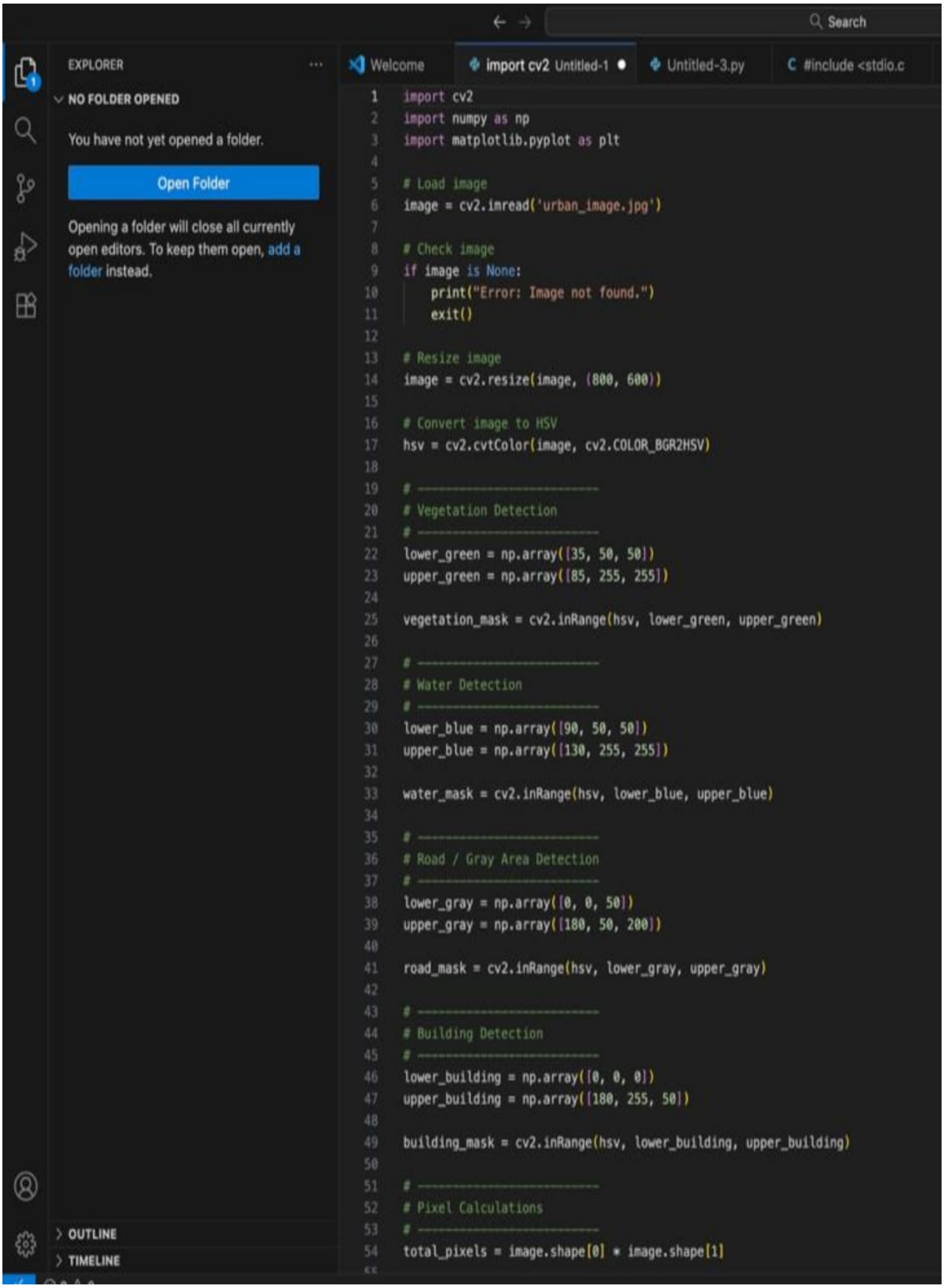
Project Solution Document

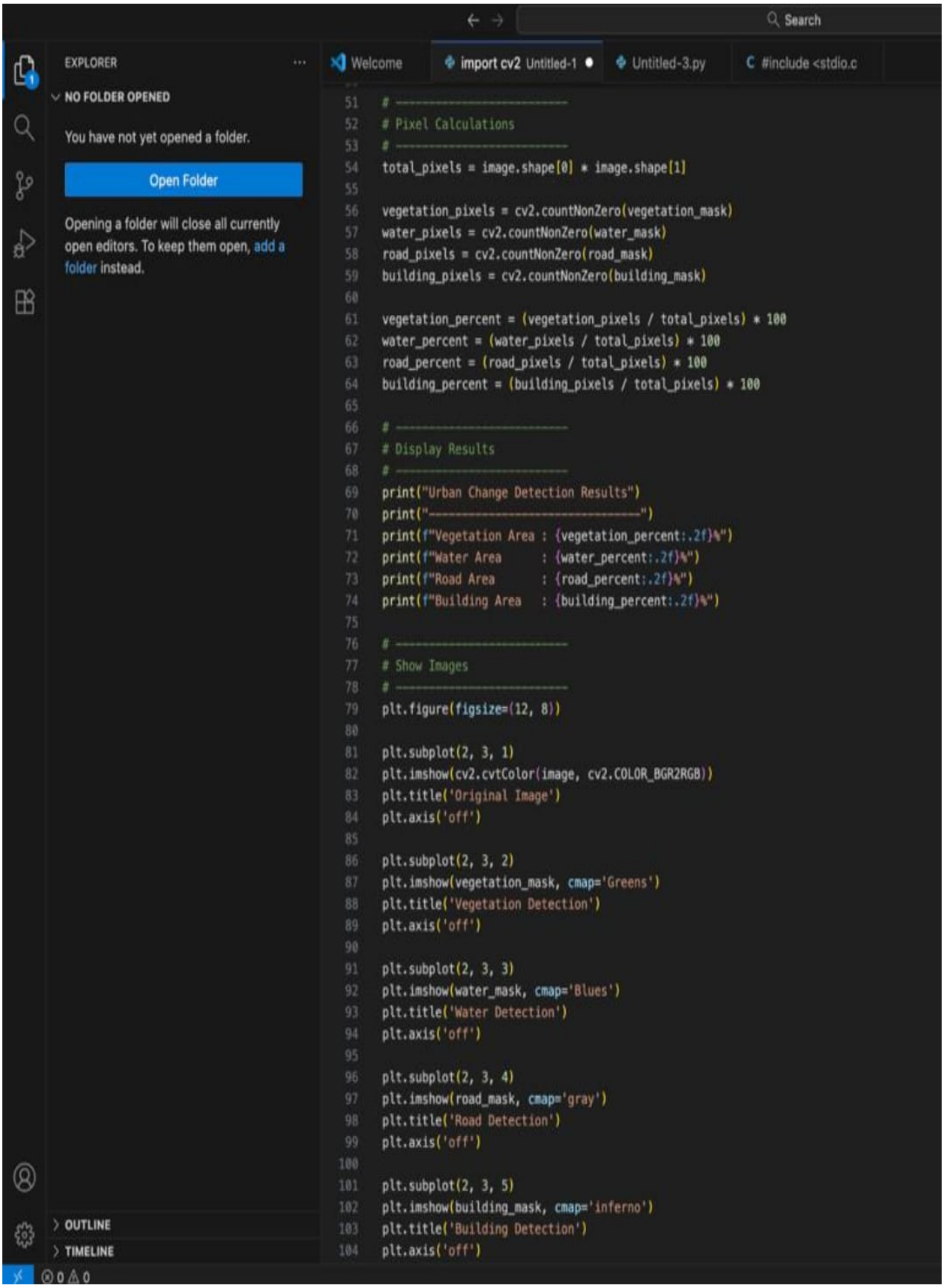
The project uses OpenCV image processing techniques for segmentation, classification, and urban change detection.

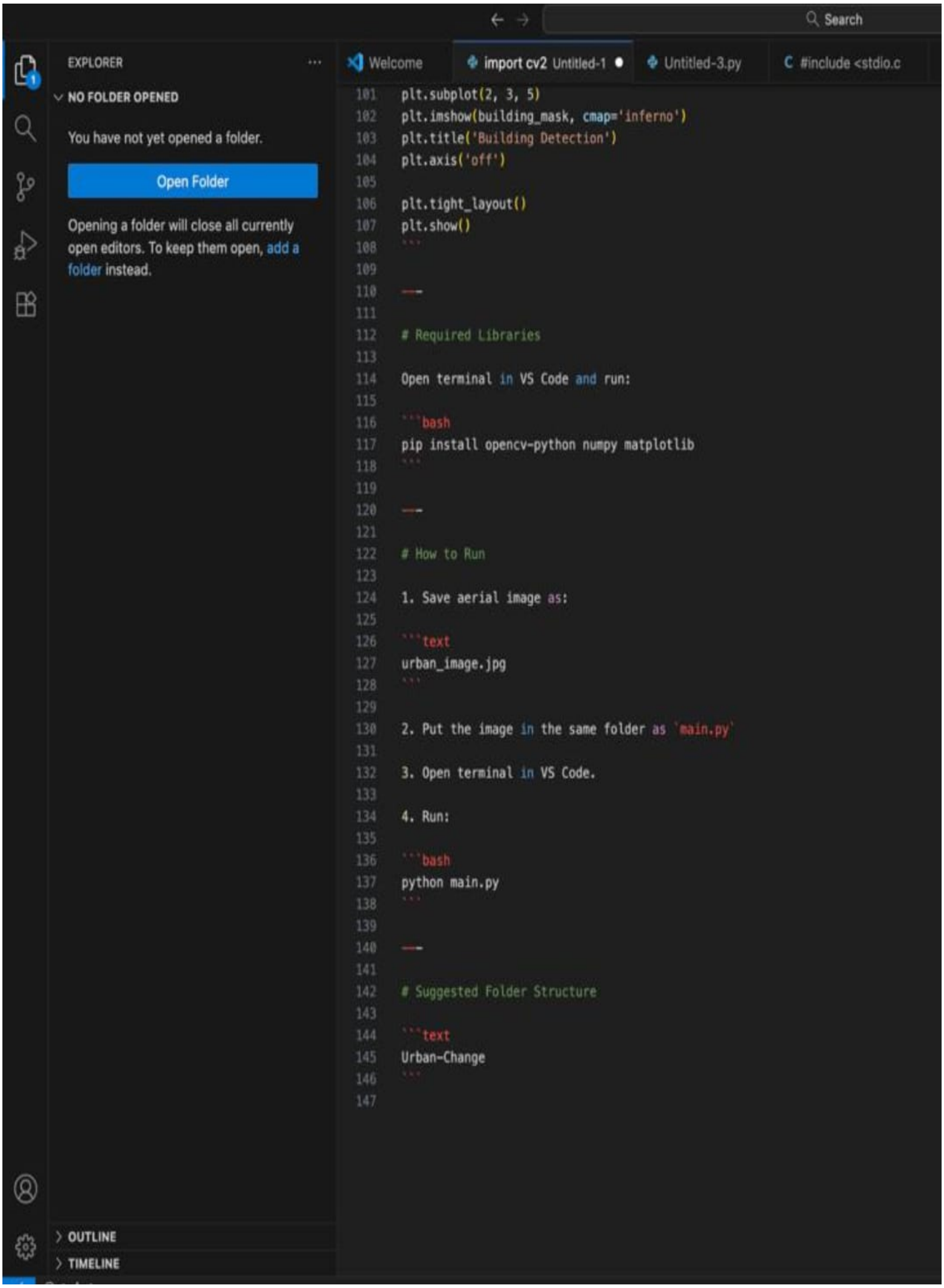
Tools and Technologies Used

Tool	Purpose
Python	Programming
OpenCV	Image Processing
NumPy	Calculations
Matplotlib	Visualization
VS Code	Development

Python Program Implementation


Search
EXPLORER
NO FOLDER OPENED
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
OUTLINE
TIMELINE
Welcome
import cv2 Untitled-1
Untitled-3.py
#include <stdio.c
import cv2
import numpy as np
import matplotlib.pyplot as plt
# Load image
image = cv2.imread('urban_image.jpg')
# Check image
if image is None:
print("Error: Image not found.")
exit()
# Resize image
image = cv2.resize(image, (800, 600))
# Convert image to HSV
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
# ------------------------
# Vegetation Detection
# ------------------------
lower_green = np.array([35, 50, 50])
upper_green = np.array([85, 255, 255])
vegetation_mask = cv2.inRange(hsv, lower_green, upper_green)
# ------------------------
# Water Detection
# ------------------------
lower_blue = np.array([90, 50, 50])
upper_blue = np.array([130, 255, 255])
water_mask = cv2.inRange(hsv, lower_blue, upper_blue)
# ------------------------
# Road / Gray Area Detection
# ------------------------
lower_gray = np.array([0, 0, 50])
upper_gray = np.array([180, 50, 200])
road_mask = cv2.inRange(hsv, lower_gray, upper_gray)
# ------------------------
# Building Detection
# ------------------------
lower_building = np.array([0, 0, 0])
upper_building = np.array([180, 255, 50])
building_mask = cv2.inRange(hsv, lower_building, upper_building)
# ------------------------
# Pixel Calculations
# ------------------------
total_pixels = image.shape[0] * image.shape[1]


Search
EXPLORER
NO FOLDER OPENED
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
OUTLINE
TIMELINE
Welcome
import cv2 Untitled-1
Untitled-3.py
#include <stdio.c
# ---------------------------
# Pixel Calculations
# ---------------------------
total_pixels = image.shape[0] * image.shape[1]
vegetation_pixels = cv2.countNonZero(vegetation_mask)
water_pixels = cv2.countNonZero(water_mask)
road_pixels = cv2.countNonZero(road_mask)
building_pixels = cv2.countNonZero(building_mask)
vegetation_percent = (vegetation_pixels / total_pixels) * 100
water_percent = (water_pixels / total_pixels) * 100
road_percent = (road_pixels / total_pixels) * 100
building_percent = (building_pixels / total_pixels) * 100
# ---------------------------
# Display Results
# ---------------------------
print("Urban Change Detection Results")
print("-----------------------------------")
print(f"Vegetation Area : {vegetation_percent:.2f}%")
print(f"Water Area   : {water_percent:.2f}%")
print(f"Road Area   : {road_percent:.2f}%")
print(f"Building Area   : {building_percent:.2f}%")
# ---------------------------
# Show Images
# ---------------------------
plt.figure(figsize=(12, 8))
plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
plt.subplot(2, 3, 2)
plt.imshow(vegetation_mask, cmap='Greens')
plt.title('Vegetation Detection')
plt.axis('off')
plt.subplot(2, 3, 3)
plt.imshow(water_mask, cmap='Blues')
plt.title('Water Detection')
plt.axis('off')
plt.subplot(2, 3, 4)
plt.imshow(road_mask, cmap='gray')
plt.title('Road Detection')
plt.axis('off')
plt.subplot(2, 3, 5)
plt.imshow(building_mask, cmap='inferno')
plt.title('Building Detection')
plt.axis('off')



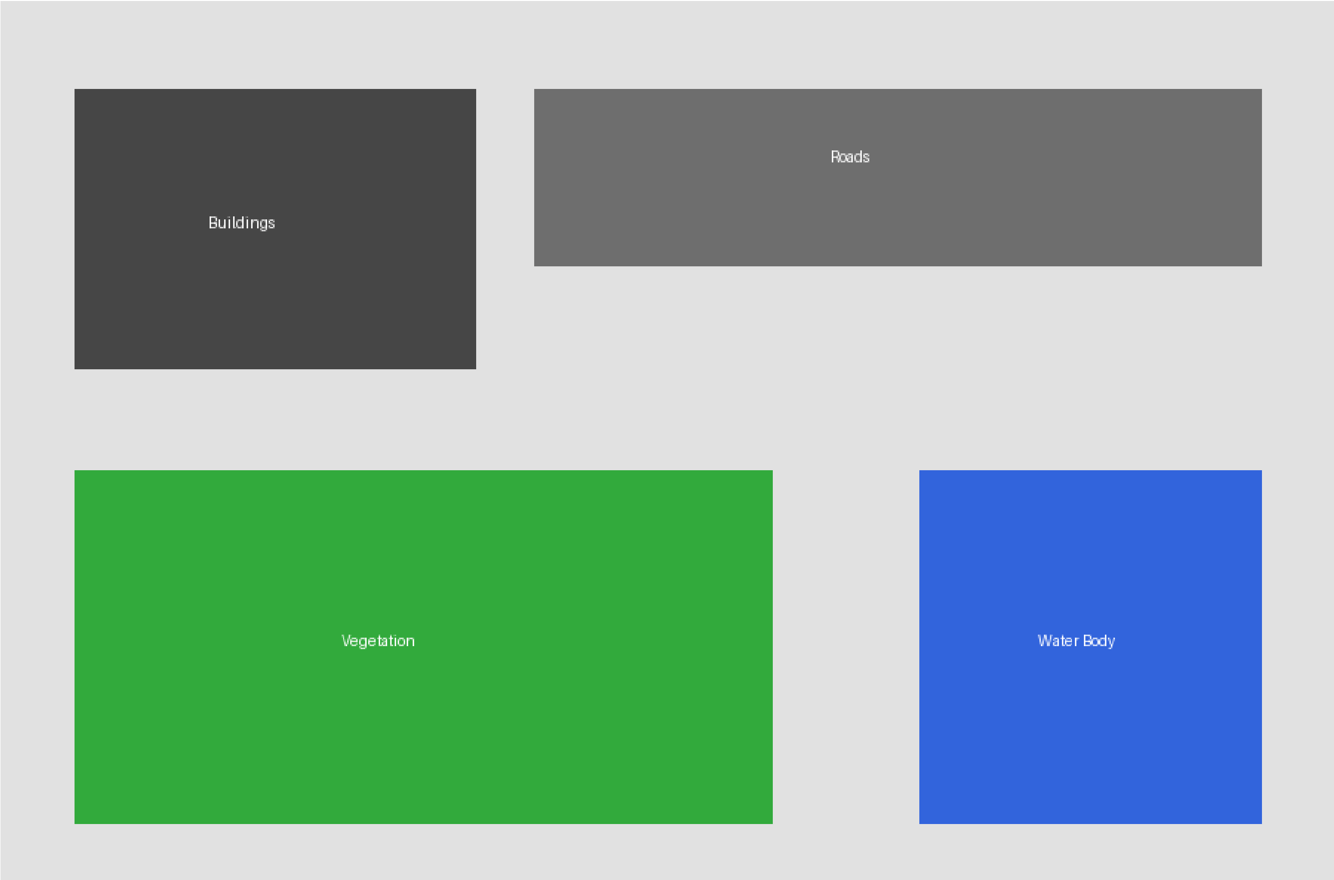
Search
EXPLORER
NO FOLDER OPENED
You have not yet opened a folder.
Open Folder
Opening a folder will close all currently open editors. To keep them open, add a folder instead.
Welcome
import cv2 Untitled-1
Untitled-3.py
#include <stdio.c
plt.subplot(2, 3, 5)
plt.imshow(building_mask, cmap='inferno')
plt.title('Building Detection')
plt.axis('off')
plt.tight_layout()
plt.show()
```
---
# Required Libraries
Open terminal in VS Code and run:
```bash
pip install opencv-python numpy matplotlib
```

How to Run
1. Save aerial image as:
```text
urban_image.jpg
```
2. Put the image in the same folder as `main.py`
3. Open terminal in VS Code.
4. Run:
```bash
python main.py
```

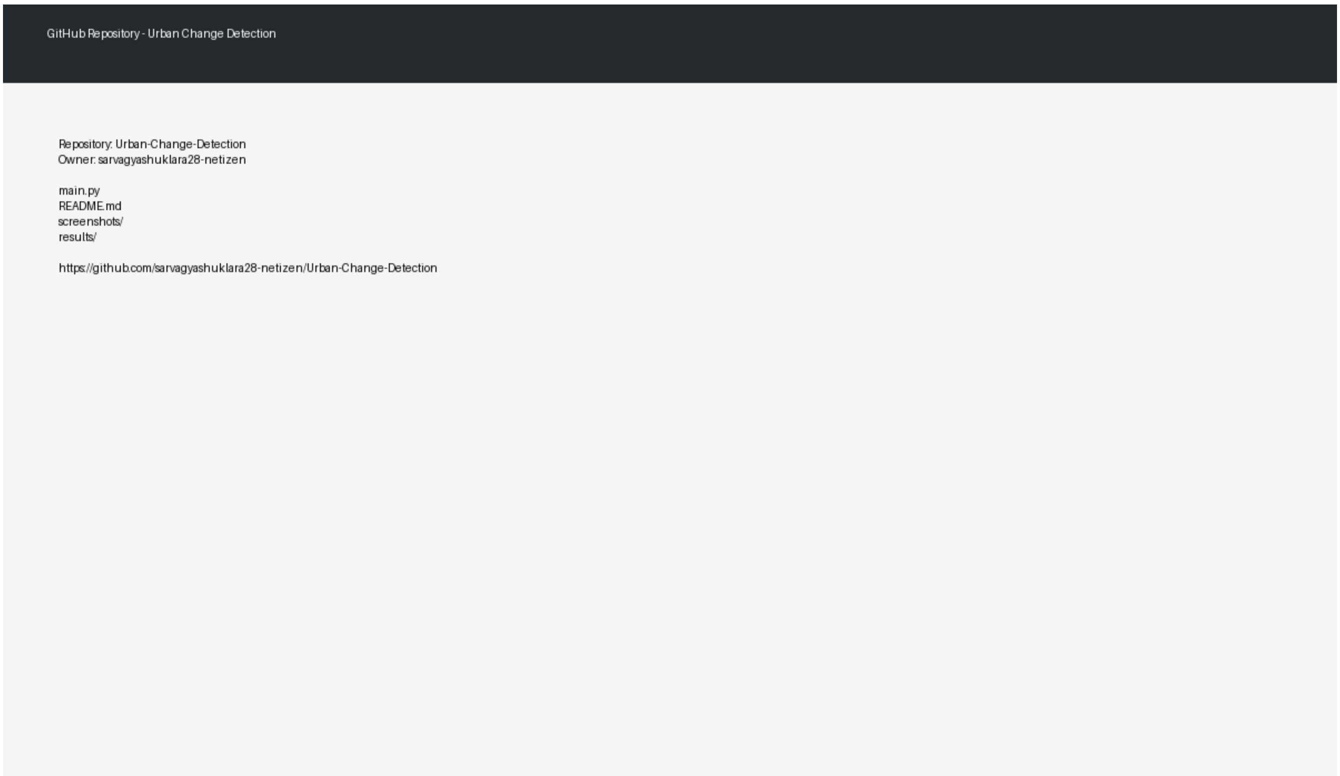
Suggested Folder Structure
```text
Urban-Change
```
OUTLINE
TIMELINE
```

Program Execution Output

Segmented Output Result



GitHub Repository Screenshot



Sample Python Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('urban_image.jpg')
hsv = cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
```

Expected Results

The system successfully classifies aerial images and calculates percentage distribution.

Conclusion

This project demonstrates urban change detection and encroachment monitoring using OpenCV.